

NAVIGATING THE UNKNOWN

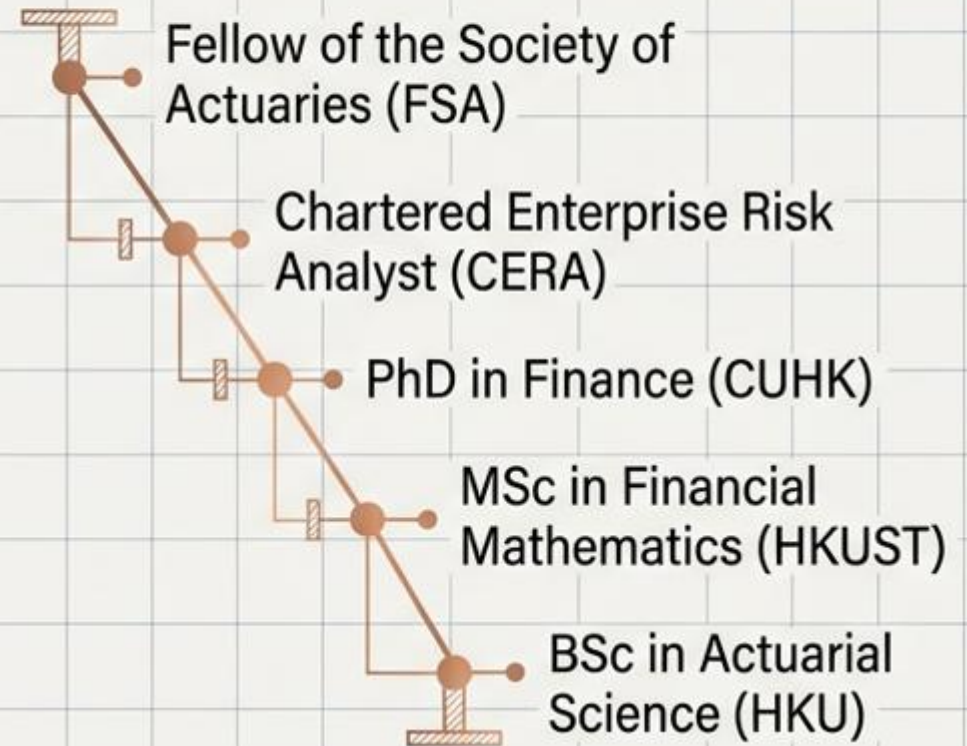
THE ACTUARIAL PROFESSION

Instructor: Dr. Derrick Fung

Leadership & Industry Roles

- Associate Dean, School of Decision Sciences (HSUHK)
- Associate Professor, Dept. of Mathematics, Statistics and Insurance
- Programme Director, MSc in Insurance
- Non-Executive Director, HKMC Annuity Limited
- Regulatory experience at the Office of the Commissioner of Insurance (2006–2014)

Academic Progression & Credentials



01



What is an Actuary?

The Professional Who Measures Risk



Definition

An actuary is a professional who uses mathematics, statistics, and financial theory to study uncertain future events — and to help organizations plan and pay for them.

Core skills:

- Mathematics & statistics
- Financial & economic theory
- Business judgment & communication



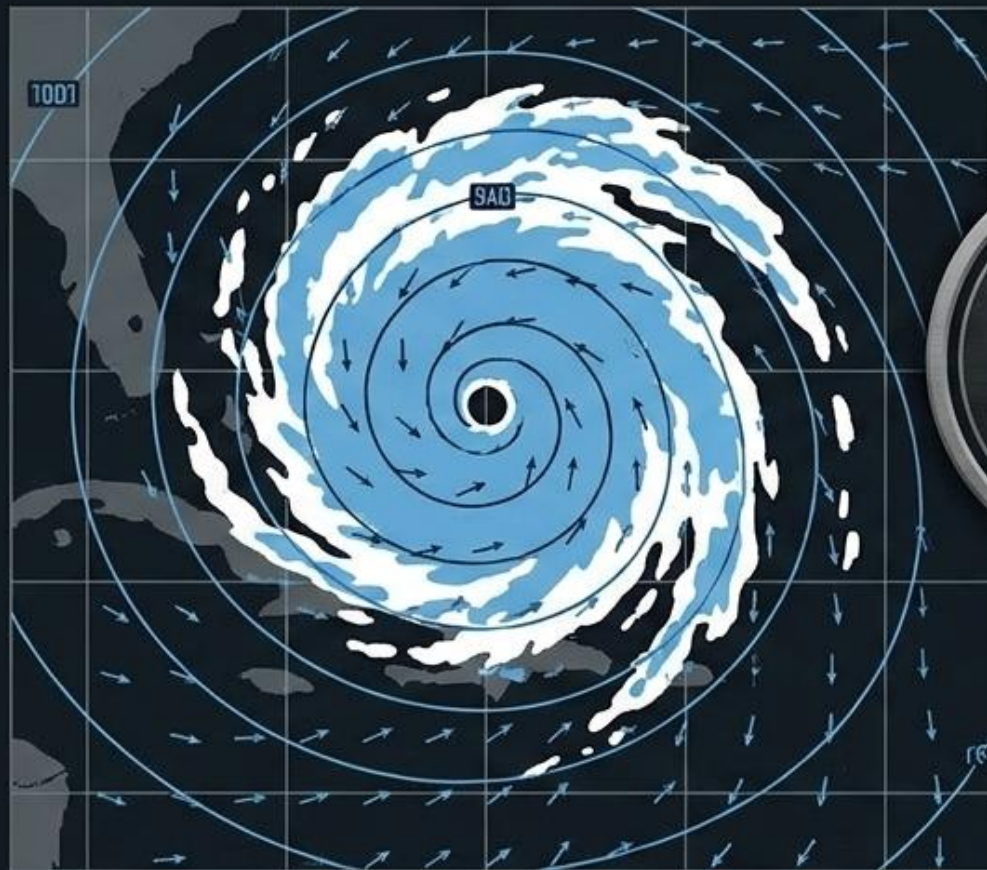
Think of it like...

“A weather forecaster — but for financial and life risks.”

Just as forecasters use data and models to predict storms, actuaries use data and models to predict things like illness, accidents, or how long people will live — then price that risk fairly.

➤ PREDICTING THE STORM

THE METEOROLOGIST



RADAR: ATLANTIC BASIN
SYSTEM STATUS: TRACKING HURRICANE

WIND SPEED: 150 MPH
PRESSURE: 940 MB

PREDICTING
THE FUTURE
USING DATA

THE ACTUARY

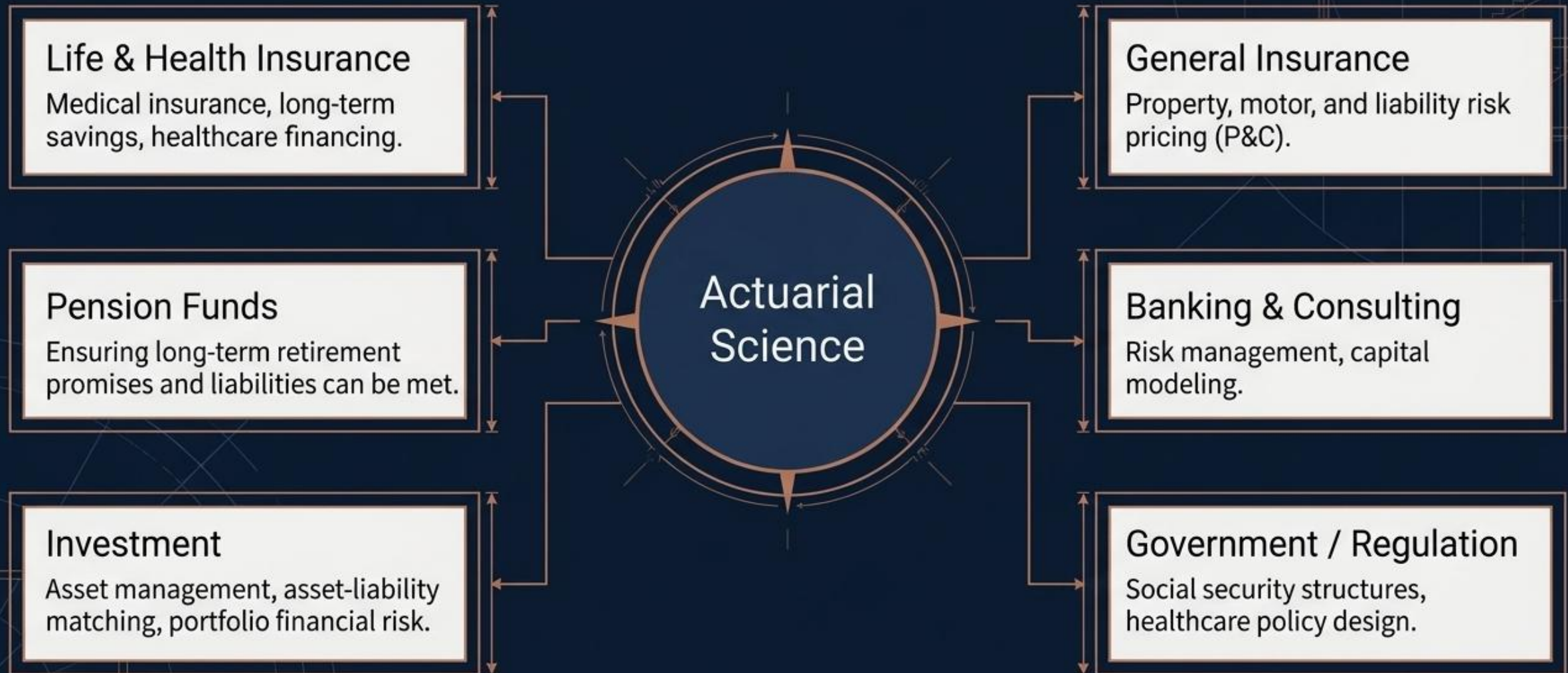


RISK DISTRIBUTION: CATASTROPHE MODELING

DATA: HISTORICAL & PREDICTIVE MODELS
EXPECTED LOSS: \$50B

PROBABILITY OF EXCEEDANCE: 1%

Actuarial skills translate across diverse financial sectors



THE ROLES OF ACTUARY IN THE INSURANCE SECTOR



- **Pricing**

Actuaries help determine the appropriate premiums for insurance policies based on the assessed risks. They use statistical models and calculations to ensure that premiums accurately reflect the expected claims and expenses.

- **Reserving**

Actuaries are responsible for estimating and setting aside reserves to cover potential future claims and liabilities. They analyze historical data, claims experience, and other factors to determine the appropriate amount of reserves.

THE ROLES OF ACTUARY IN THE INSURANCE SECTOR



- **Financial Modeling and Forecasting**

Actuaries develop financial models and forecasts to assess the financial health of insurance companies. They analyze various scenarios and evaluate the impact of different factors on the company's financial performance.

- **Product Development**

Actuaries play a key role in developing new insurance products and modifying existing ones. They analyze market trends, customer behavior, and risk factors to design insurance products that are competitive, profitable, and meet regulatory requirements.

THE ROLES OF ACTUARY IN THE INSURANCE SECTOR



- **Regulatory Compliance**

Actuaries ensure that insurance companies comply with regulatory requirements and standards. They provide expertise in areas such as solvency, financial reporting, and capital adequacy.

- **Risk Management**

Actuaries help insurance companies identify, assess, and manage risks. They assess and quantify risks associated with insurance policies, including analyzing historical data, evaluating potential future events, and determining the likelihood of claims and losses.

MINI-BOSS CHALLENGE: THE PRICING PUZZLE

THE SCENARIO:

You are insuring 100 people.

THE RISK:

Statistically, exactly 1 person will file a claim for a broken leg. The hospital bill is \$10,000.

THE QUESTION:

How much premium must you charge all 100 people to exactly cover the risk?



ANSWER:



-
- What could go wrong with poor actuarial knowledge?



CASE FILE 01: THE DOMINO EFFECT (HIH INSURANCE)

THE EVENT

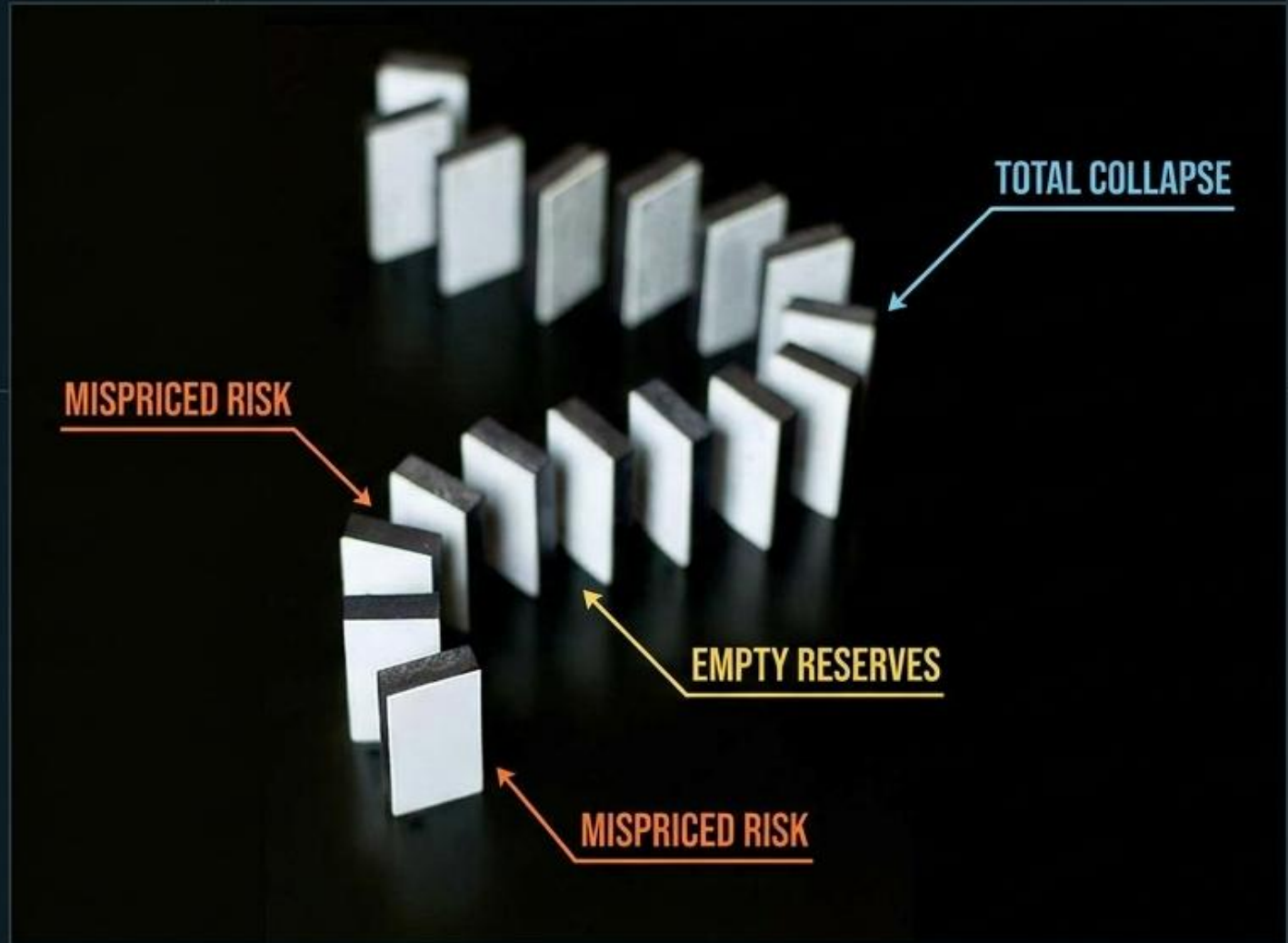
HIH Insurance collapsed in 2001.

THE FATAL ERROR

Aggressive expansion and mispriced liability insurance. The company underestimated the risks.

THE RESULT

Inadequate reserves. They simply did not set aside enough funds to meet their obligations to people who needed payouts.



CASE FILE 02: THE 50% WIPEOUT (EQUITABLE LIFE)

THE PROMISE:

Equitable Life offered policies with Guaranteed Annuity Rates, promising a fixed 7% return upon retirement.

THE BLIND SPOT:

Poor actuarial support failed to account for the costs if interest rates fell.

THE FALLOUT:

Interest rates dropped, creating a massive shortfall. The company nearly collapsed.



-50%

Retirement wipeout for policyholders.

02



Actuary Ranking

How the profession compares to other careers

A consistently top-ranked career built on stability and growth

U.S. News & World Report (2026)

#5

Best
Technology
Jobs

#7

Best
STEM Jobs

#11

in 100 Best
Jobs overall

The Actuarial Advantage



Strong starting and
long-term salary.



Low physical risk combined
with steady demand.



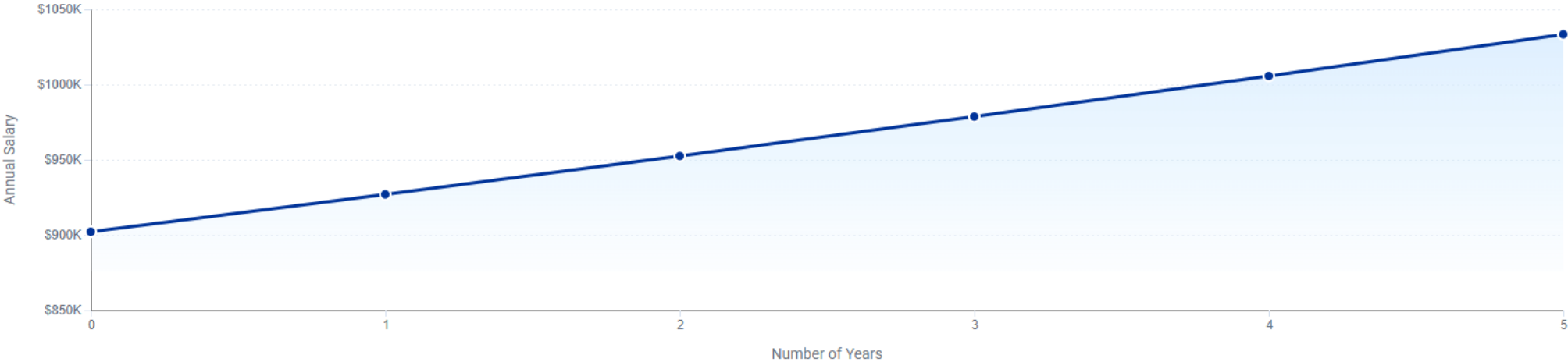
Clear, predictable
credential-based progression.



High job security across
varying economic cycles.

Salary Potential for an Actuary (Fellow)

Explore the estimated salary growth for an Actuary (Fellow) in Hong Kong SAR for up to 5 years. The estimated salary of an Actuary (Fellow) will be HK\$926,963 in 1 year and HK\$1,033,374 in 5 years.



View Salary by Year
2 Years ▾

Estimated Salary
HK\$952.5K

Average Growth
5.6%

The estimated salary for an Actuary (Fellow) in 2 years is HK\$952,492, which is 5.6% average growth. This information is based on ERI's compensation database.

The optimal balance of reward, effort, and lifestyle

Profession	Entry Path Length	Relative Stress Level	Work-Life Balance
Actuary	3-4 yr degree + exams (done part-time while working)	 (Moderate)	 (Good)
Doctor	5-6 yr degree + years of residency	 (High)	 (Limited)
Lawyer	3-4 yr degree + professional qualification	 (High)	 (Limited)
Engineer	3-4 yr degree	 (Moderate)	 (Good)
Data Scientist	3-4 yr degree (+ master's often required)	 (Moderate)	 (Good)

 High Stress / Good Balance  Moderate Stress / Moderate Balance  Limited Stress / Limited Balance

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A Day in the Life

What actuaries actually do at work

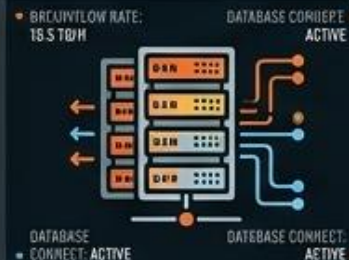


INSIDE THE CONTROL ROOM: A DAY IN THE LIFE

09:00 AM

09:00 AM | DATA GATHERING

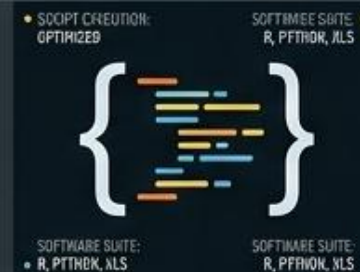
Pulling massive datasets from company databases.



11:00 AM

11:00 AM | THE TOOLKIT

Coding and crunching numbers using Excel, R, Python, and specialized software.



02:00 PM

02:00 PM | RISK MEETINGS

Translating complex math into business strategies for the CEO.



04:00 PM

04:00 PM | REPORT GENERATION

Finalizing the financial models.



Many Paths Within the Profession



Life Actuary

Life insurance & long-term
savings products



Health Actuary

Medical insurance & healthcare
financing



General Insurance

Property, motor, liability (P&C)
risk



Pension Actuary

Retirement plans & long-term
liabilities



Investment Actuary

Asset management & financial
risk

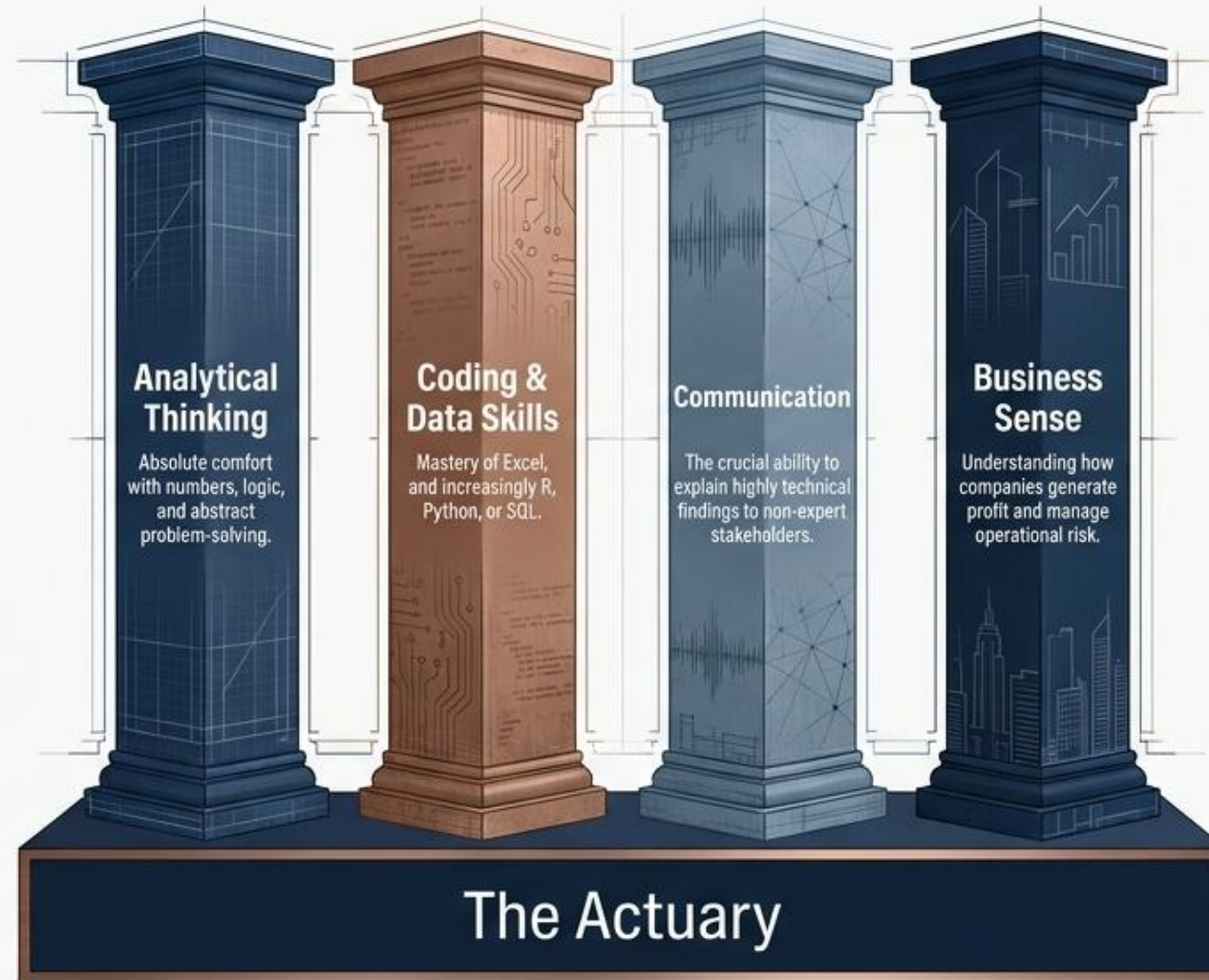
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How to Become an Actuary

The path from secondary school to qualification

The modern actuary requires a hybrid skill stack



Supported by internationally recognized professional bodies

IAA (International Actuarial Association)

The global umbrella body that sets international standards and connects all national associations.

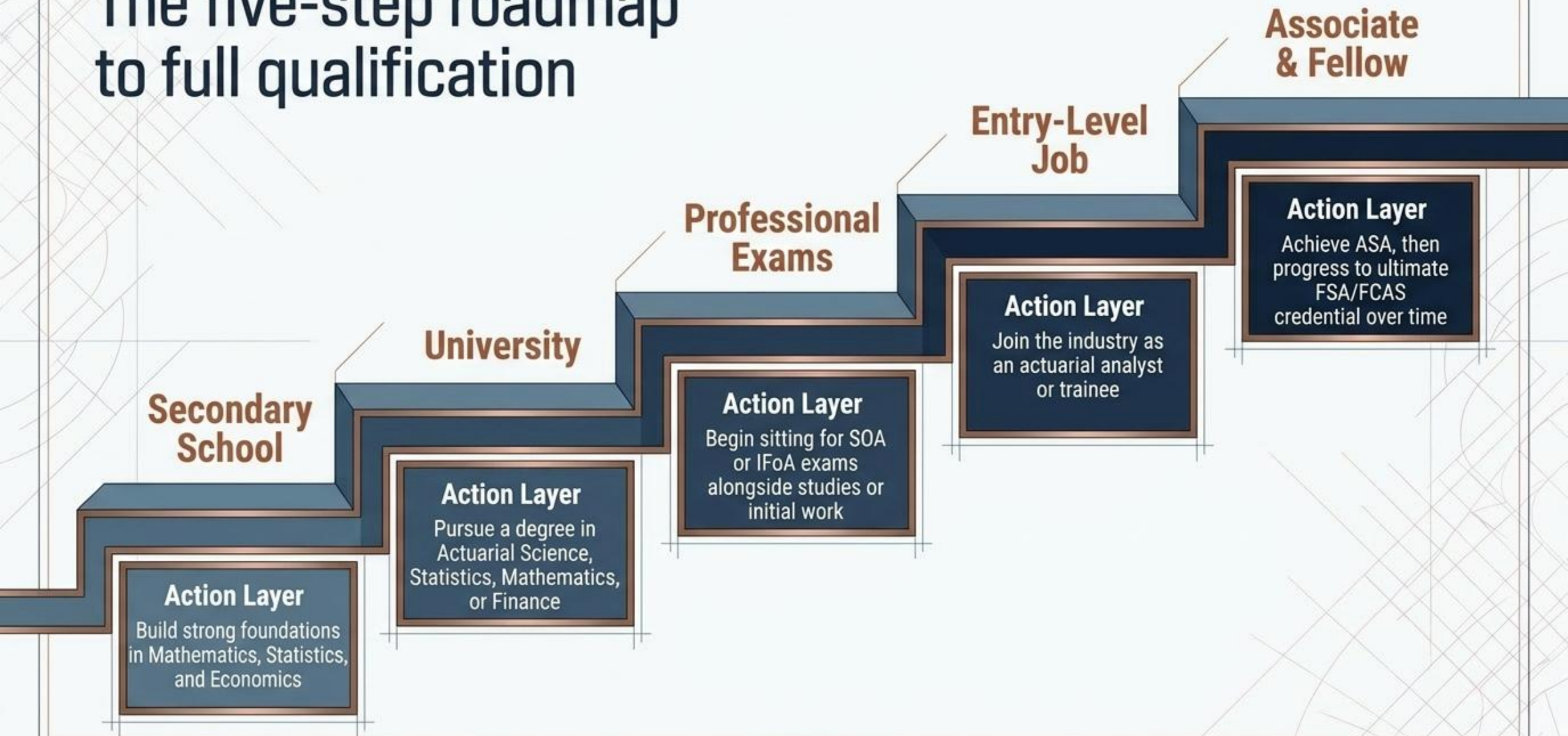
SOA (Society of Actuaries)

United States / Global focus.
Specializes in life, health, retirement, and general insurance.

IFoA (Institute and Faculty of Actuaries)

United Kingdom / Global focus.
Widely recognized and utilized in Hong Kong and across Asia.

The five-step roadmap to full qualification



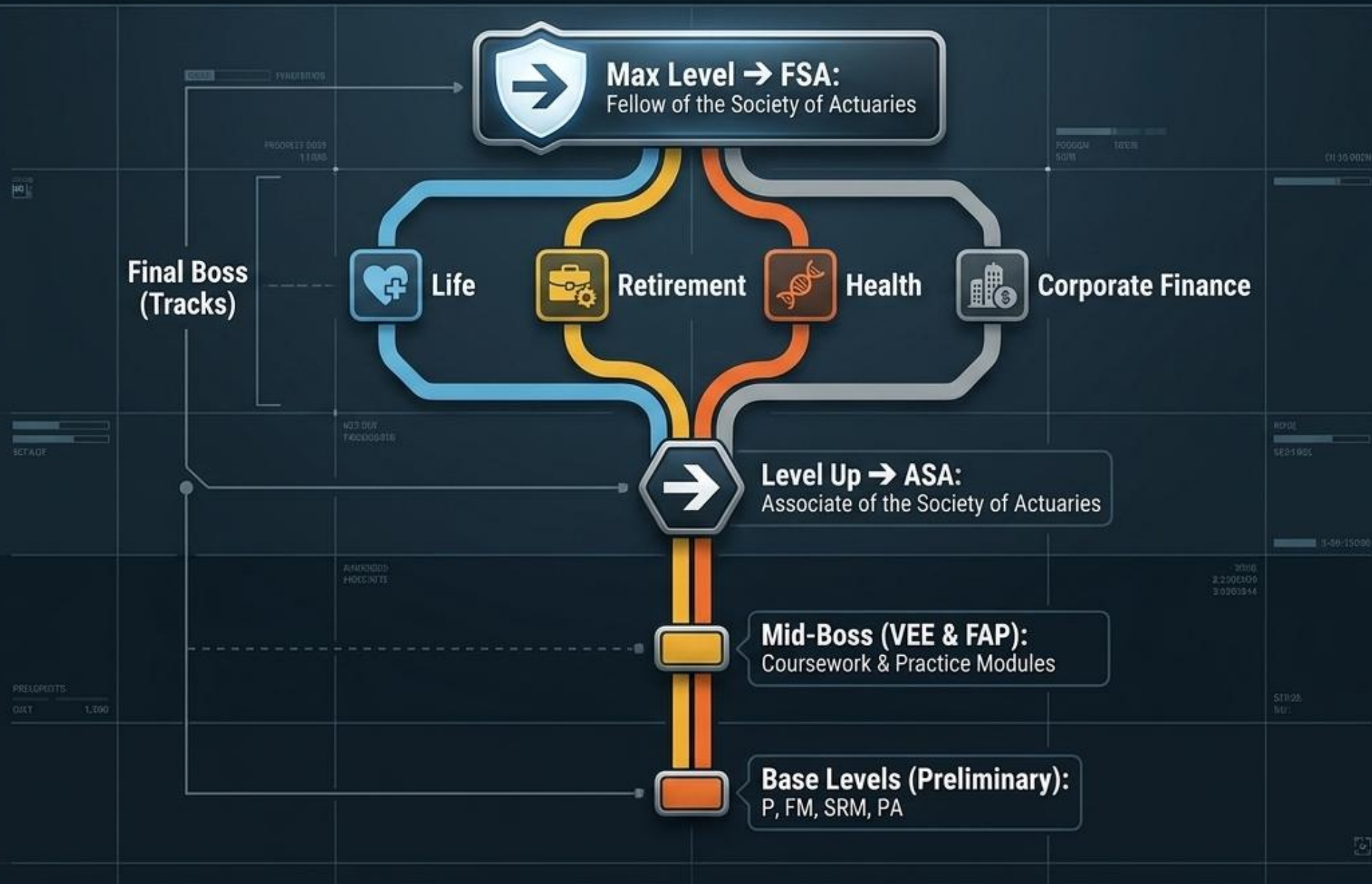
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The SOA Exam Structure

Understanding the road to full qualification

LEVELING UP: THE SOA SKILL TREE



THE REALITY CHECK: WHAT IT TAKES



5-10 Years: Average time to complete post-graduation



300 Hours: Study time per exam



High Reward: Employer-sponsored study time, massive salary bumps per exam passed, and bulletproof job security.

OPERATIONAL READINESS: REALITY CONFIRMED ●



Key Takeaways

1

Actuaries turn uncertainty into manageable numbers — a rare and valuable skill.

2

It's a highly ranked, stable career built on math, statistics, and business sense.

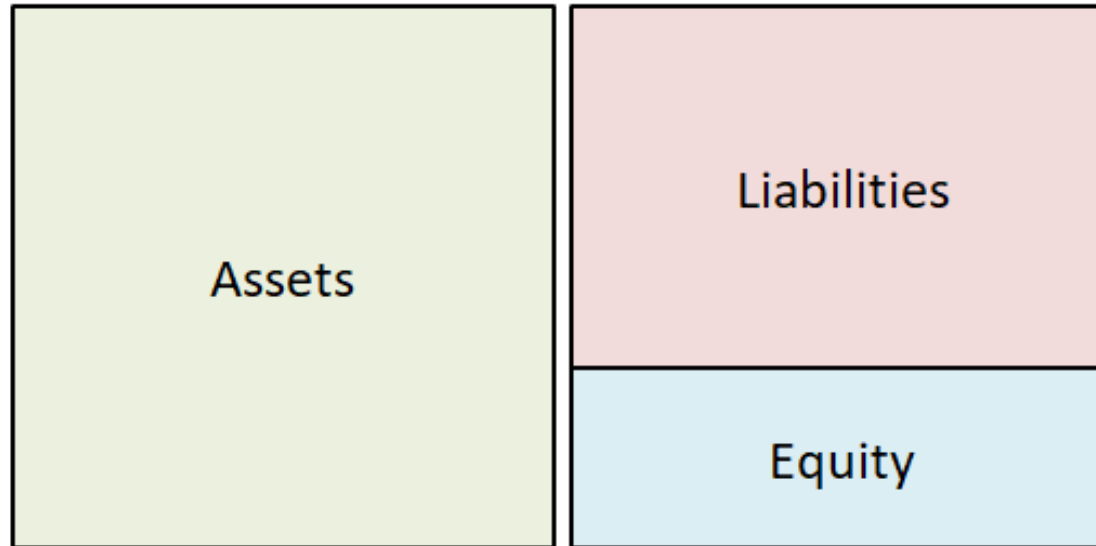
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The path starts today: strong math foundations, curiosity, and steady exam progress.

MINI BOSS CHALLENGE: WHAT HAPPENS WHEN INTEREST RATE DECREASES?

Balance Sheet Assets, Liabilities and Equity

When interest rate decreases, assets increase or decrease?



When interest rate decreases, liabilities increase or decrease?

When interest rate decreases, equity increase or decrease?

Objective: We want to minimize the fluctuation of equity to interest rate movements